

Algorithmic Choice in Bluesky

Responsible ranking + security in an open feed ecosystem



Measured: discovery → panel → 600,651 ranked impressions (Feb 1-2, 2026)

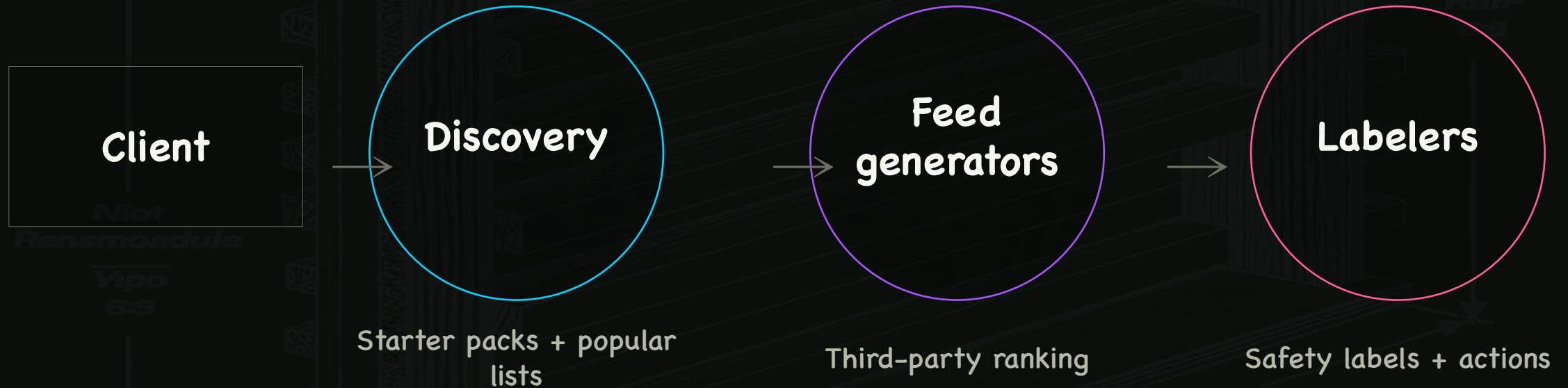
Algorithmic choice $\neq$ decentralized power

- Feeds + labelers turn ranking/moderation into a marketplace.
- Discovery + hosting create chokepoints + new attack surfaces.

Thesis: pluralism shifts power to discovery + infrastructure.

Bluesky's open feed ecosystem

Ranking + moderation become third-party services (a marketplace).



Chokepoints emerge where users find feeds + who hosts them. [ICWSM'24 10.1609/icwsml.31293]

Discovery is measurable (not anecdotes)

Starter packs + popular feeds are explicit lists that seed attention.

Starter packs (onboarding bundles)

Starter pack (metadata)

Name	Creator DID	Pack URI
Lauren Raymor's Starter Pack	did:plc:dwzj...b7caw1	.../app.bsky.graph.starterpack/3mdto57g

Example row (truncated)

Starter pack → feed inclusion

Pack	Feed
.../app.bsky.graph.starterpack/3mdto57gqjd2g	.../app.bsky.feed.generator/mich-dot-gov

Example row (truncated)

Popular feeds (in-app list)

Popular feeds (example row)

Rank	Feed	URI
top	Discover	.../app.bsky.feed.generator/whats-hot

Example row (truncated)

These surfaces seed attention → panel sampling. [ICWSM'25 10.1609/icwsm.v19i1.35810]

What we measured (H1-H6)

Discovery → provider leverage → exposure → overlap → rich-get-richer → safety labels

H1 Discovery concentration

Do a few feeds dominate starter-pack inclusion?



H2 Provider leverage

Which providers get outsized discovery?



H3 Exposure concentration

How concentrated is exposure?



H4 Feed overlap

Do feeds show the same posts?



H5 Exposure vs author size

Do larger authors get outsized exposure?



H6 Risk exposure variability (labels)



Problem: responsible ranking meets open markets

- Users choose algorithms – but discovery chooses defaults.
- Responsible ranking needs: security, fairness, safety, accountability.

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Ransmoedule

Vipo
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Glacled
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Threat model: marketplace adversaries

- Malicious feeds/rankers can steer exposure or surface explicit content.
- Growth gaming + label gaps enable evasion, bias, and targeting.

[NDSS'21 10.14722/ndss.2021.24525; USENIX Sec'23 10.5555/3620237.3620333]

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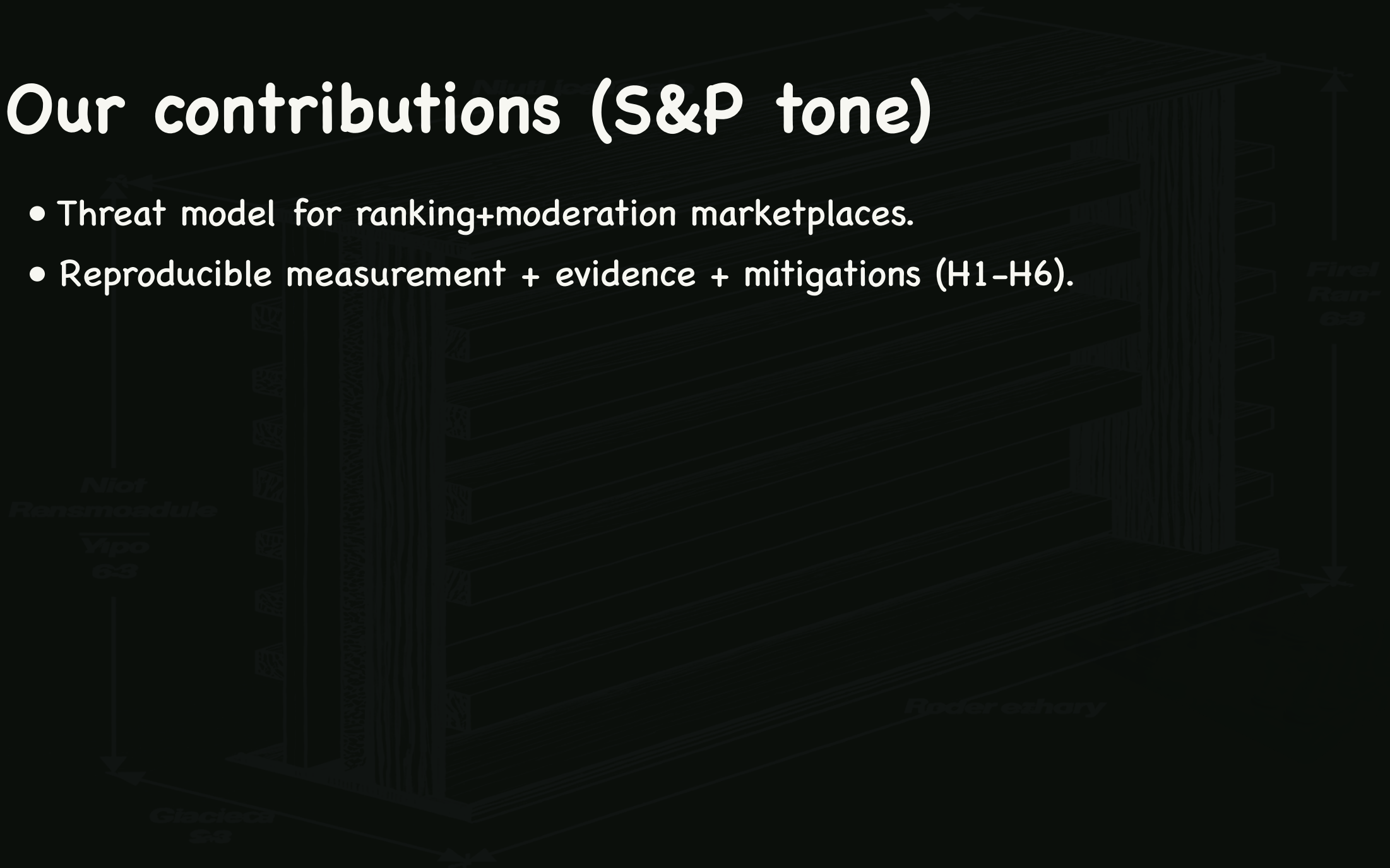
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Our contributions (S&P tone)

- Threat model for ranking+moderation marketplaces.
- Reproducible measurement + evidence + mitigations (H1-H6).



Methods credibility (read-only snapshot)

Discovery → fixed panel → ranked impressions → joinable tables

1

Relay scan

Enumerate candidate actors

2

Feed index

Hydrate feed generators

3

Starter packs

Extract discovery signals

4

Popular

Capture ranking surface

Read-only XRPC to Relay/AppView (only POST is session auth/refresh).

The run behind this story

Run settings (minimal)

viewer: unauth (public AppView)
window: 2026-02-01T09:04:54Z → 2026-02-02T08:23:24Z
run_id: 88b597ae...bd1967

Anchors (2 numbers)

Panel size: 6,000 feeds
Captured: 600,651 ranked impressions

Every stage leaves a trace (exports)

- Feed generator index
- Starter packs + inclusions
- Popular feed list
- Feed panel
- Ranked impressions
- Posts (metadata) + authors

Feed generator index

Feed	Provider	URI
Discover	discover.bsky.app	.../app.bsky.feed.generator/whats-hot

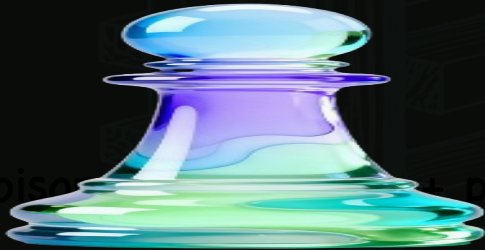
Example row (truncated)

Keys line up: feed_uri / post_cid / author_did

Adversary playbook: 4 ways markets get gamed

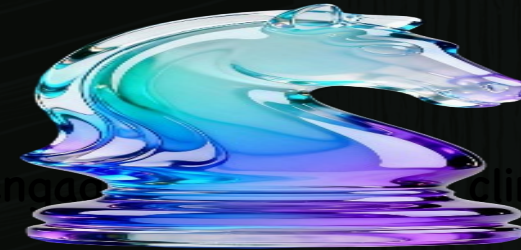
Market incentives create predictable attack patterns.

Discovery poisoning



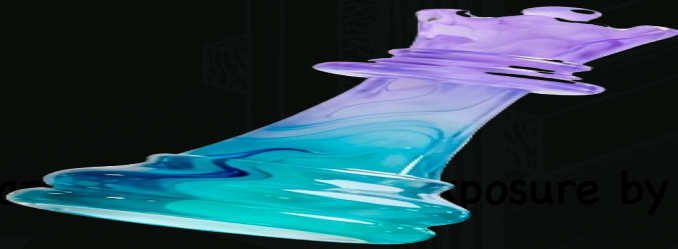
Capture/poison top popular lists.

Growth gaming



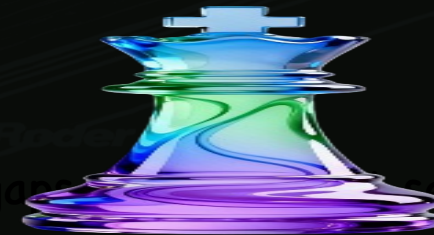
Sybil + engagement to climb ranks.

Malicious rankers



3rd-party exposure by design.

Label gaps/evasion



Coverage gaps in semantics.

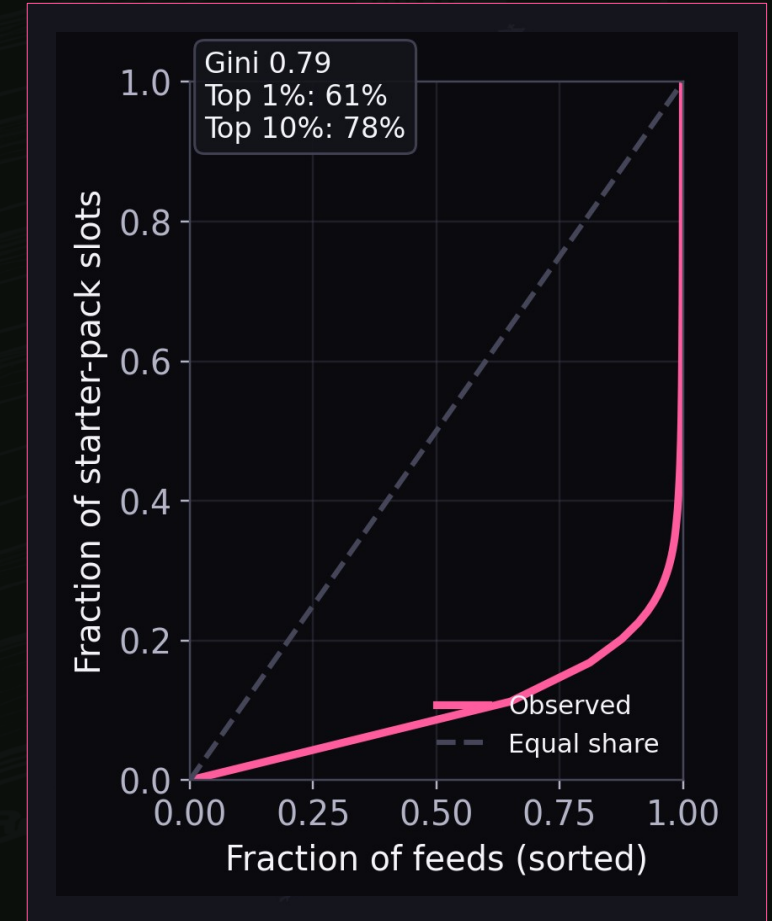
H1 - Discovery slots are concentrated

Top 1% share of starter-pack slots



61%

Defaults become gatekeepers → cheaper discovery capture.



H2 – Discovery creates provider leverage

Max leverage: discovery share vs
hosting footprint



x801

Hosting chokepoints become targets →
ecosystem-wide risk.



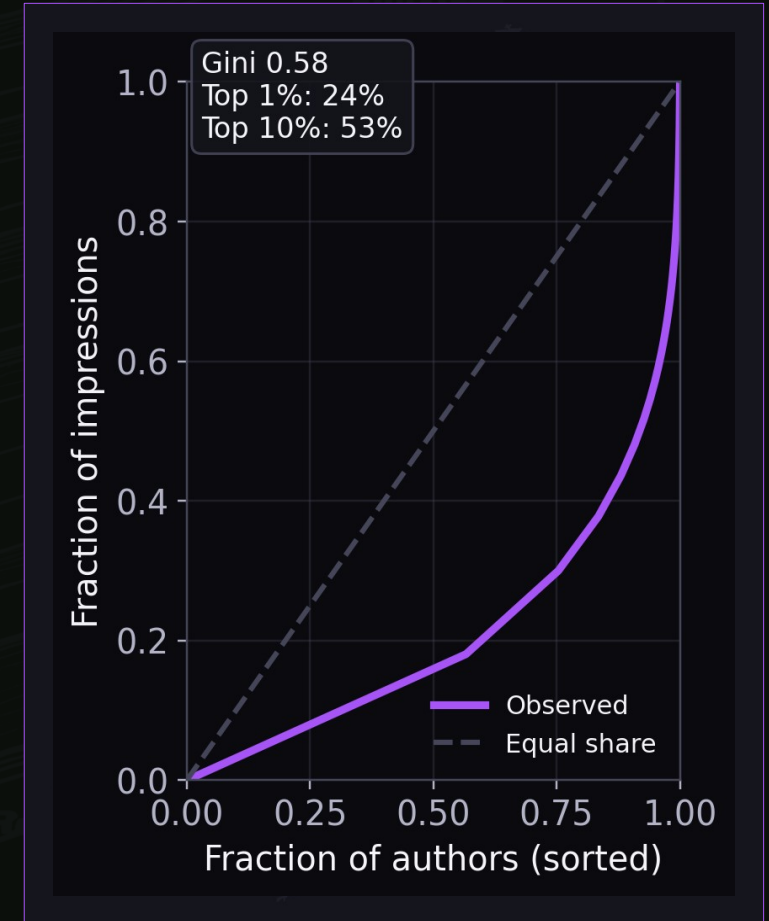
H3 - Exposure is concentrated among authors

Top 10% of authors get this share of impressions



53%

Concentrated exposure lowers manipulation/targeting cost.



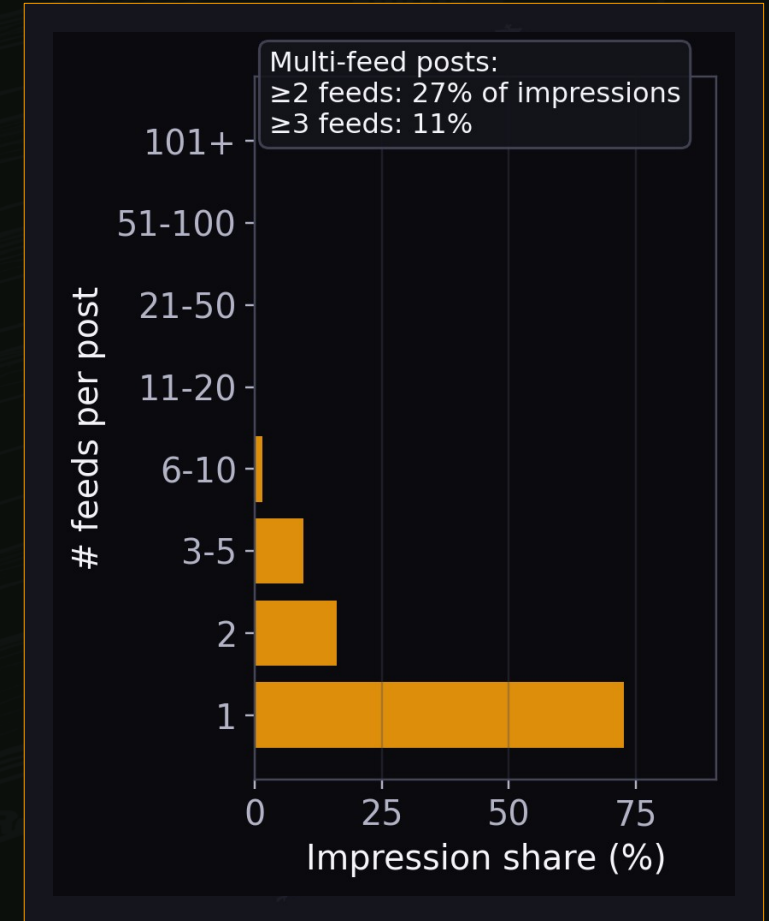
H4 – Many feeds show the same posts

Share of impressions for posts in ≥ 2 feeds



27%

Overlap amplifies winners $\rightarrow$ pluralism without diversity.



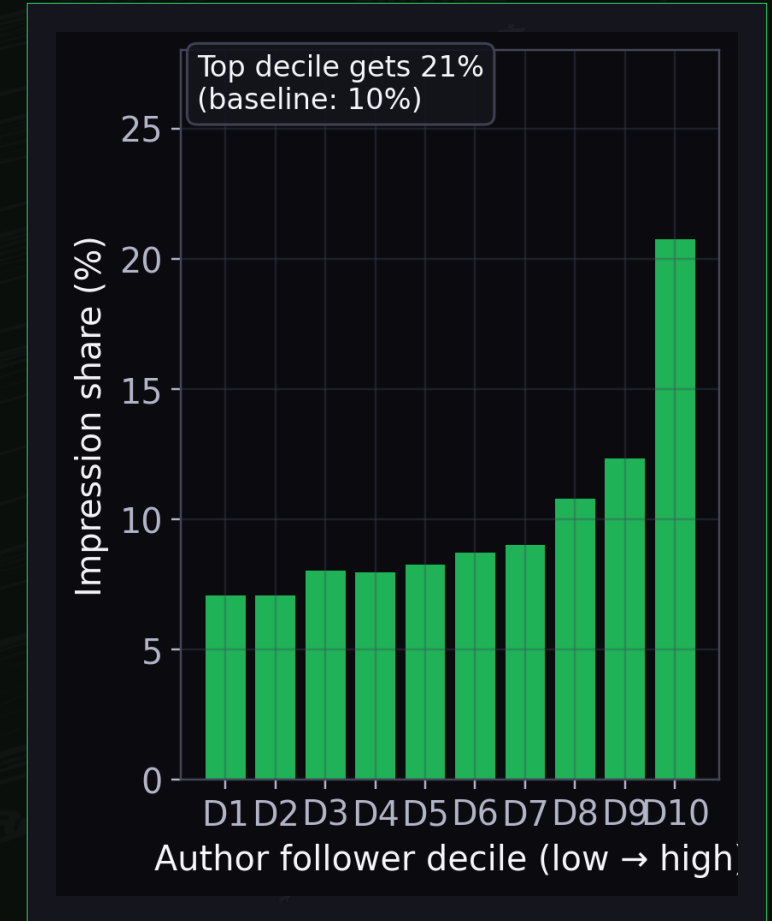
H5 – Bigger accounts get outsized exposure

Top follower decile gets this share of impressions



21%

Popularity bias reappears → unfair + gameable outcomes.



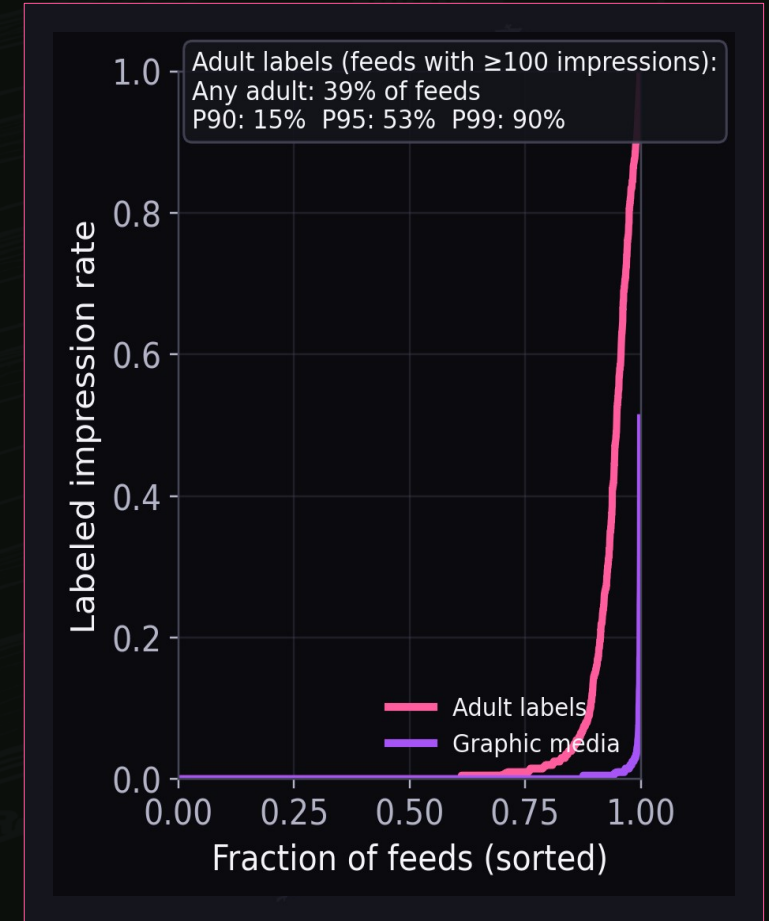
H6 - Safety label exposure varies by feed

Feeds (≥ 100 impressions) with any adult labels



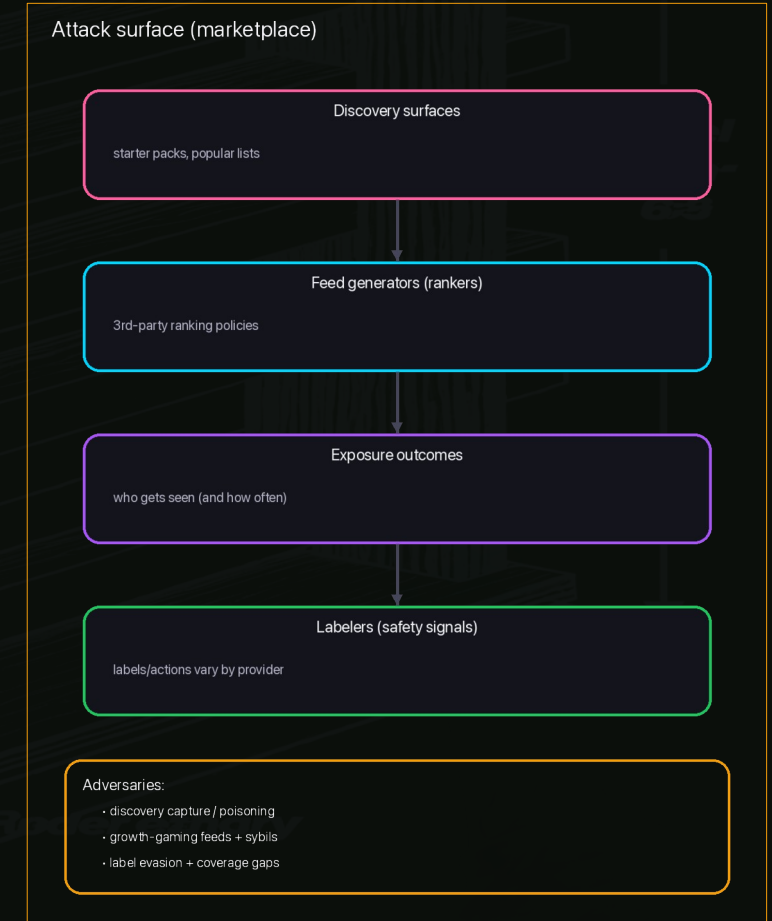
39%

Safety boundaries depend on feed choice
→ uneven risk.



Security implications: a new attack surface

- Concentrated discovery reduces attack cost.
- 3rd-party rankers can be adversarial → marketplace-wide audit + guardrails.



Privacy: third-party services can profile

Request patterns + outputs can reveal preferences and identity.

GET

`https://.../xrpc/app.bsky.feed.getFeed?feed=at://...&cursor=...`

What a service sees

- IP + timing + feed URI (and viewer DID if auth)
- repeated queries + outputs reveal preferences

Why it matters

- profiling / tracking + output leakage
- cross-service linkage risk

[CCS'21 10.1145/3460120.3484770]

Implication: decentralized moderation is fragile

- Label semantics are subjective → disagreement across labelers.
- Feed choice can bypass/amplify safety boundaries (coverage gaps + bias).

[NAACL'22 10.18653/v1/2022.naacl-main.431; Findings ACL'22 10.18653/v1/2022.findings-acl.176]

Mitigation: discovery transparency + diversification

- Publish logs for starter packs + popular lists.
- Diversify defaults + detect manipulation (growth + overlap spikes).

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Mitigation: auditable feed marketplace

Logs + reproducible snapshots enable accountability

Add audit hooks around discovery, ranking, and labelers.

Expose stable identities + policies so clients can warn and researchers can audit.

Transparency logs

Audit APIs

Reproducible snapshots

Why this matters

Choice only helps if users can compare risk, diversity, and trust.

What to disclose (minimal)

- provider identity + hosting footprint
- discovery surfaces (who promoted what)
- ranking output summaries (exposure / overlap)
- labeler policies + coverage
- incident response + takedown hooks

Mitigation: “feed nutrition labels” (user-facing)

- Show provider, overlap, and exposure concentration at a glance.
- Show label exposure + applied labelers (warnings/filters/defaults).



Mitigation: labeler interoperability + secure sharing

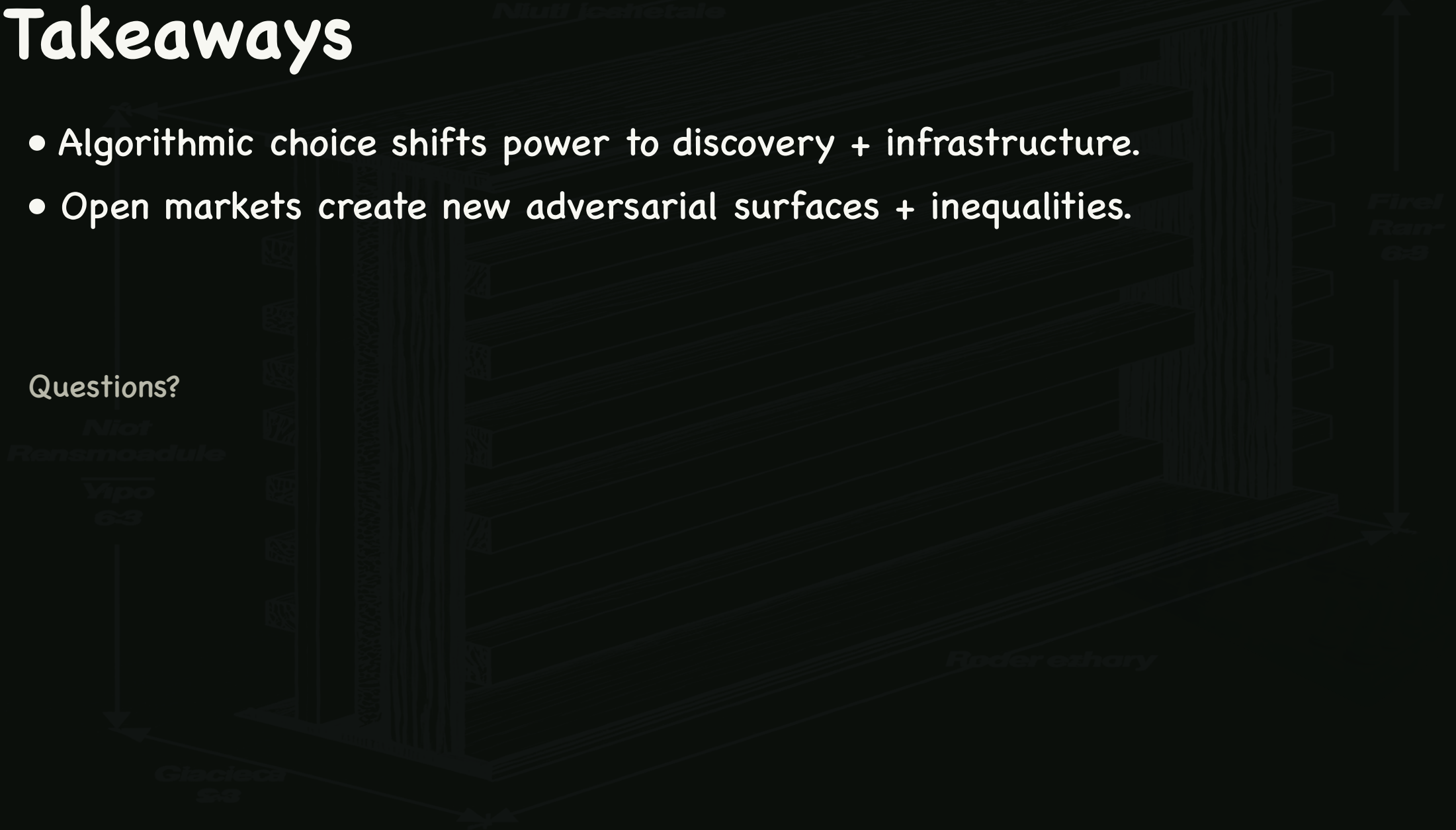
- Standardize vocabularies + transparency for labeler policies.
- Secure/private sharing + multi-labeler consensus to resist poisoning.

[IEEE S&P'22 10.1109/SP46214.2022.9833647]

Takeaways

- Algorithmic choice shifts power to discovery + infrastructure.
- Open markets create new adversarial surfaces + inequalities.

Questions?



References (venue-filtered, 2021-2025)

CCS / IEEE S&P / NDSS / USENIX Sec / WWW / CSCW / ICWSM / ACL / KDD

Decentralization + discovery

ICWSM'25 Starter Packs
<https://doi.org/10.1609/icwsm.v19i1.35810>

ICWSM'24 Decentralised Moderation
<https://doi.org/10.1609/icwsm.v18i1.31293>

ICWSM'24 Group Sanctions
<https://doi.org/10.1609/icwsm.v18i1.31316>

WWW'23 Will Admins Cope?
<https://doi.org/10.1145/3543507.3583487>

CSCW'24 Trouble in Paradise
<https://doi.org/10.1145/3687059>

Ranking + security + labels

NDSS'21 Poisoning Recommenders
<https://doi.org/10.14722/ndss.2021.24525>

USENIX'23 PORE
<https://doi.org/10.5555/3620237.3620333>

CCS'21 Membership Inference
<https://doi.org/10.1145/3460120.3484770>

IEEE S&P'22 FL Poisoning Eval
<https://doi.org/10.1109/SP46214.2022.9833647>

KDD'21 Popularity Bias (Dynamic)
<https://doi.org/10.1145/3447548.3467376>

NAACL'22 Annotators w/ Attitudes
<https://doi.org/10.18653/v1/2022.naacl-main.431>

Findings ACL'22 Fairness Varies
<https://doi.org/10.18653/v1/2022.findings-acl.176>

Findings EMNLP'23 Hate Bias (Causal)
<https://doi.org/10.18653/v1/2023.findings-emnlp.440>